Qualcomm Technologies, Inc.

SW6100 Arm® Cortex® -M55 Zephyr RTOS Boot Flow Technical Overview

80-74841-125 Rev. AA

Revision History

Revision	Date	Description
AA	October 2025	Initial release

Contents

Introduction

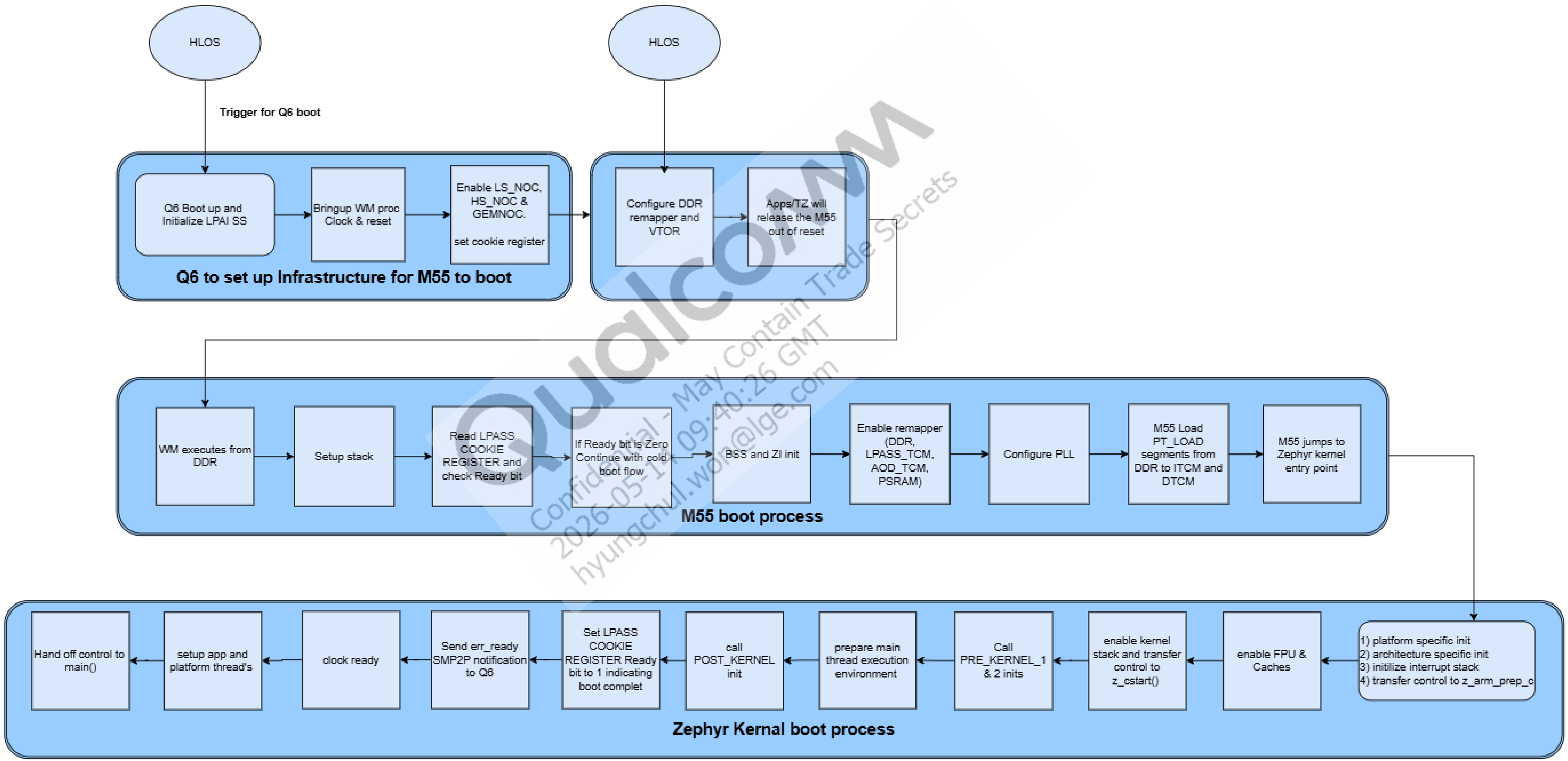
- This document provides an overview of the Zephyr RTOS boot flow in Arm® Cortex ® -M55.
- Zephyr is a small real-time operating system (RTOS) for connected resource-constrained and embedded devices supporting multiple architectures.
- Sensor wearable microcontroller (SWM), Ambient wearable microcontroller (AWM) Cortex -M55 cores use Zephyr kernel v3.7.0 as the underlying RTOS.
- The SW6100 chipset has two Cortex M55 processors, SWM and AWM, that are part of the low power artificial intelligence (LPAI) subsystem.
  - SWM is for sensors use cases and hosts Qualcomm sensing hub Sensor Algorithms.
  - AWM is responsible for managing Display rendering in the Ambient mode such as Always on Display (AoD) offload cases.

Section 1

M55 AWM/SWM Boot Flow

M55 AWM/SWM Boot Flow

The following figure shows the M55 AWM/SWM boot flow:



M55 AWM/SWM Boot Flow (cont.)

- ADSP Q6 must be booted up and configured to set up the required LPAI infrastructure for M55.
- M55 boots from DDR because RPROC on HLOS does not support copying TCM segments to M55's internal TCM.
- TrustZone (TZ) authenticates the M55 firmware, enables a remapper to allow M55 to access DDR, and then releases the reset.
- A tiny bootloader is implemented within the M55 firmware that runs from DDR. The vector table offset register (VTOR) points to the DDR address.
- Tiny bootloader: `wm_proc/core_ect/boot`
  - Initializes the remaining remappers—LPASS TCM, AOD TCM, AOSS RAM, and the PMIC region—to enable M55 to access external memory.
  - Configures PLL, adjusts core clock so that TCM copying can be done.
  - Loads segments from DDR to data tightly coupled memory (DTCM) and instruction tightly coupled memory (ITCM).
  - Relocates the VTOR to TCM vector table and jumps to the Zephyr entry function defined in the linker script.

`wm_proc/core_ect/boot/subsys_boot/subsys_boot.c`

```
/* Execute entry point */
ctxt->entry_fn();
```

`zephyr/kernel/config/soc/arm/aspen_lpai/linker.ld :: ENTRY(CONFIG_KERNEL_ENTRY)`

`zephyr/kernel/config/aop.conf::  CONFIG_KERNEL_ENTRY="Reset_Handler" (core_ect/boot/arch/cortex_m/startup.c)`

- From the Zephyr kernel entry point, M55 begins execution from its internal TCM.
- Subsequently, M55 initializes the platform and the interrupt stack.

`wm_proc/zephyr/zephyr/arch/arm/core/cortex_m/vector_table.h`

M55 AWM/SWM Boot Flow (cont.)

- Transfers control to `z_prep_c.c` at `wm_proc/zephyr/zephyr/arch/arm/core/cortex_m/prep_c.c`. This enables the FPU and caches.
- Next, transfers control to `z_cstart()` at `wm_proc/zephyr/zephyr/kernel/init.c`.

```
enum init_level {
     INIT_LEVEL_EARLY = 0,
     INIT_LEVEL_PRE_KERNEL_1,
     INIT_LEVEL_PRE_KERNEL_2,
     INIT_LEVEL_POST_KERNEL,
     INIT_LEVEL_APPLICATION,
  #ifdef CONFIG_SMP
     INIT_LEVEL_SMP,
  #endif /* CONFIG_SMP */
  };
```

- `z_cstart()` calls `INIT_LEVEL_PRE_KERNEL_1` and `INIT_LEVEL_PRE_KERNEL_2` sys inits.
- It then calls `bg_thread_main()`, which does `INIT_LEVEL_POST_KERNEL` sys init and `INIT_LEVEL_APPLICATION` sys init.

M55 AWM/SWM Boot Flow (cont.)

- Error-ready-irq is done as part of `INIT_LEVEL_POST_KERNEL` in `err_init()` located at

  `wm_proc/core_ect/debugtools/err/src/err.c`.

- This marks the end of the boot flow.

- The list of sys init tasks is auto-generated in the build at the following location for SWM:

  `wm_proc/config/bsp/lpai_swm_img/build/aspen.swm.prod/LPAI_SWM_IMG_aspen.swm.prod_sys_init_report.txt`

Section 2

Remappers

Remappers

- Memory remapping is a technique used to ensure that all physical memory in the system is accessible to the cores (AWM/SWM), especially in systems with large RAM.
- During the DDR boot, memory remapping is done by the bus interface unit using memory remappers located on the AXI and QHB paths.
- AWM/SWM could access DDR, LPASS_TCM, AOD_TCM, or PSRAM through the AXI/QHB interface. To access the appropriate memory, respective remapper must be configured and enabled using the device tree located at `wm_proc/core_ect/dt_settings/soc/aspen_lpai_swm/core/boot/remapper`.
- AXI remapper is for memory (DDR, LPASS_TCM, AOD_TCM, and PSRAM).
- QHB remapper is for IO mapped memory.
- Remappers configurations are not recommended to be modified by customers as they are tightly coupled with SMMU stage-2 mapping and access control mechanism.

Section 3

References

Documents and Standards

Title	Number
Qualcomm Technologies, Inc.	
SW6100 Software User Guide	80-74841-50

Acronyms and Terms

Acronym or Term	Definition
AWM	Ambient wearable microcontroller
AoD	Always on Display
DTCM	Data tightly coupled memory
ITCM	Instruction tightly coupled memory
LPAI	Low power artificial intelligence
SWM	Sensor wearable microcontroller
TZ	TrustZone
VTOR	Vector table offset register

Questions?

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